PARISHRAM 2.0

CLASS 12TH 2026

HUMAN REPRODUCTION

ZOOLOGY | LECTURE-7

-PARAM SIR



TODAY'S TOPIC -

1. PREGNANCY AND EMBRYONIC DEVELOPMENT
2. REVISION

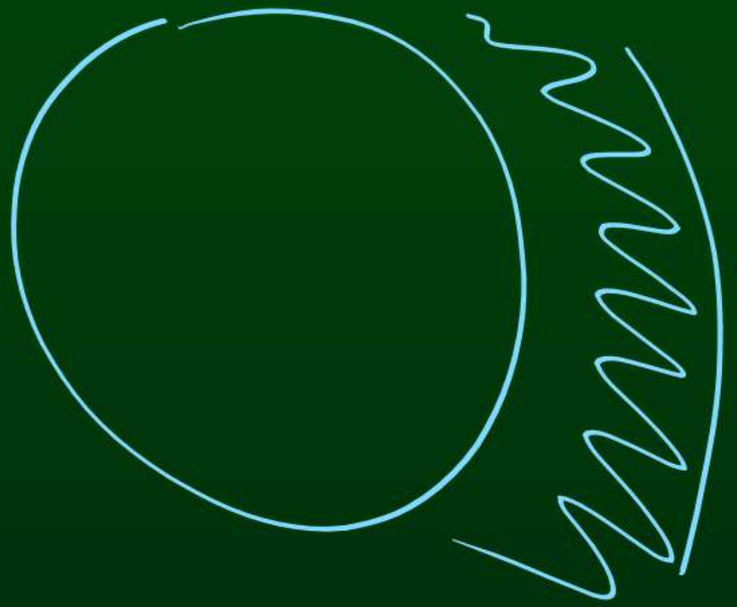


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After implantation, finger-like projections appear on the trophoblast called **chorionic villi** which are surrounded by the uterine tissue and maternal blood. The chorionic villi and uterine tissue become interdigitated with each other and jointly form a structural and functional unit between developing embryo (foetus) and maternal body called **placenta** (Figure 2.12).

The placenta facilitate the supply of oxygen and nutrients to the embryo and also removal of carbon dioxide and excretory/waste materials produced by the embryo. The placenta is connected to the embryo through an umbilical cord which helps in the transport of substances to and from the embryo. Placenta also acts as an endocrine tissue and produces several hormones like **human chorionic gonadotropin** (hCG), **human placental lactogen** (hPL), **estrogens, progestogens,** etc. In the later phase of pregnancy, a hormone called **relaxin** is also secreted by

EXTRACT



NCERT

the ovary. Let us remember that hCG, hPL and relaxin are produced in women only during pregnancy. In addition, during pregnancy the levels of other hormones like estrogens, progestogens, cortisol, prolactin, thyroxine, etc., are increased several-folds in the maternal blood. Increased production of these hormones is essential for supporting the fetal growth, metabolic changes in the mother and maintenance of pregnancy.

Immediately after implantation, the inner cell mass (embryo) differentiates

Est
Progestron
thyroxine

Store Nitrogen Waste

Endometrium

Allentois

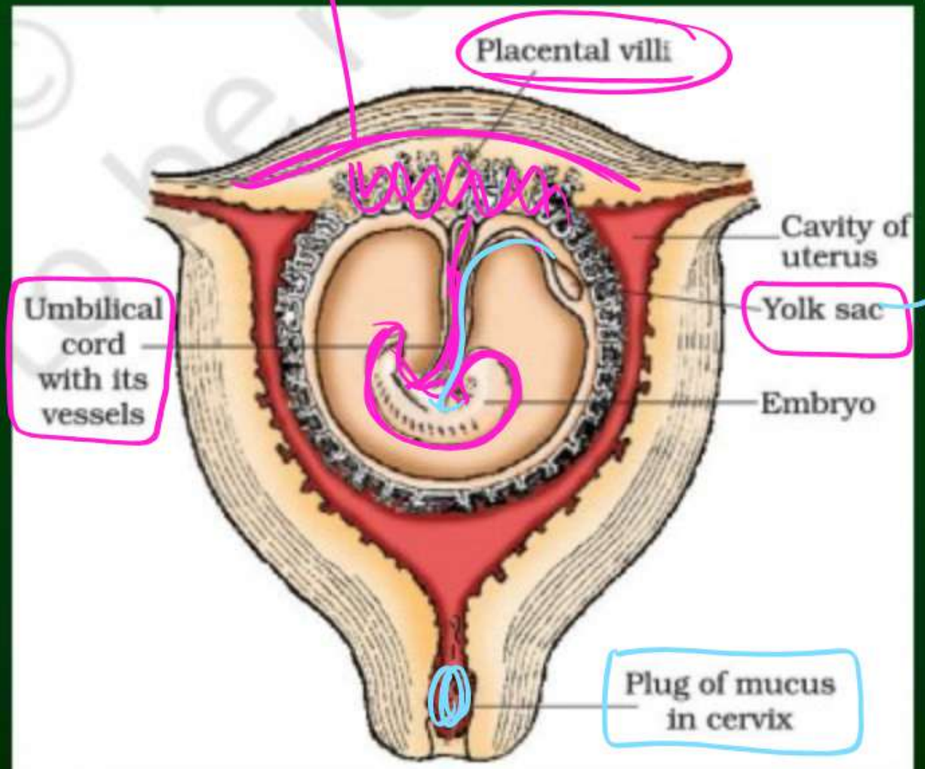


Figure 2.12 The human foetus within the uterus

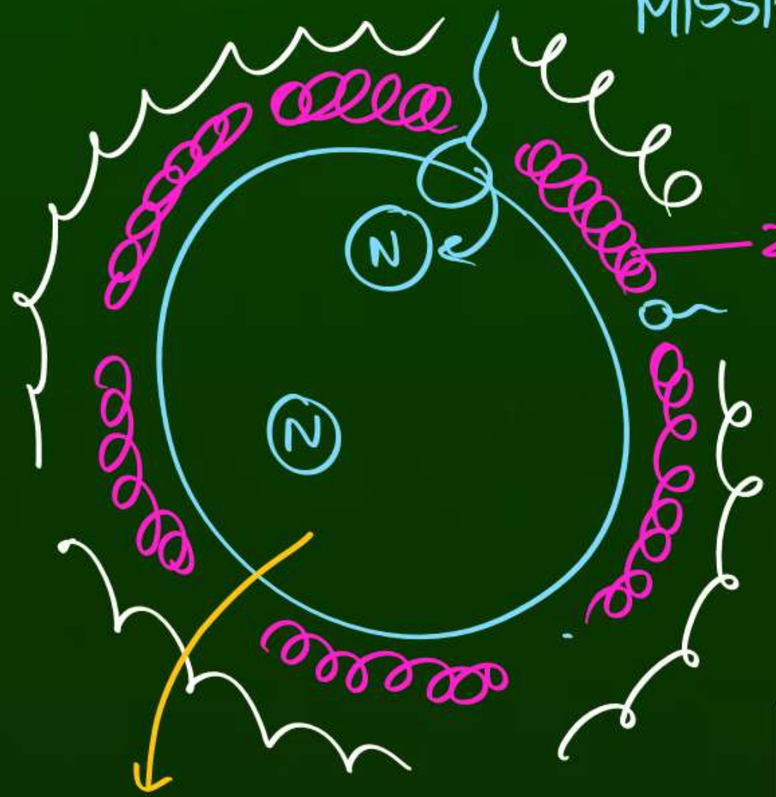
NCERT

EXTRACT

into an outer layer called **ectoderm** and an inner layer called **endoderm**. A **mesoderm** soon appears between the ectoderm and the endoderm. These three layers give rise to all tissues (organs) in adults. It needs to be mentioned here that the inner cell mass contains certain cells called **stem** cells which have the potency to give rise to all the tissues and organs.

What are the major features of embryonic development at various months of pregnancy? The human pregnancy lasts 9 months. *Do you know for how many months pregnancy last in dogs, elephants, cats?* Find out. In human beings, after one month of pregnancy, the embryo's heart is formed. The first sign of growing foetus may be noticed by listening to the heart sound carefully through the stethoscope. By the end of the second month of pregnancy, the foetus develops limbs and digits. By the end of 12 weeks (first trimester), most of the major organ systems are formed, for example, the limbs and external genital organs are well-developed. The first movements of the foetus and appearance of hair on the head are usually observed during the fifth month. By the end of about 24 weeks (end of second trimester), the body is covered with fine hair, eye-lids separate, and eyelashes are formed. By the end of nine months of pregnancy, the foetus is fully developed and is ready for delivery.

MISSION SUCCESSFUL



ZYGOTE

① ZP Role

Trophoblast

Blastocyst → Implantation

ZP remove

X

Multiple Sperm Entery

BLOCK OF POLYSPERY

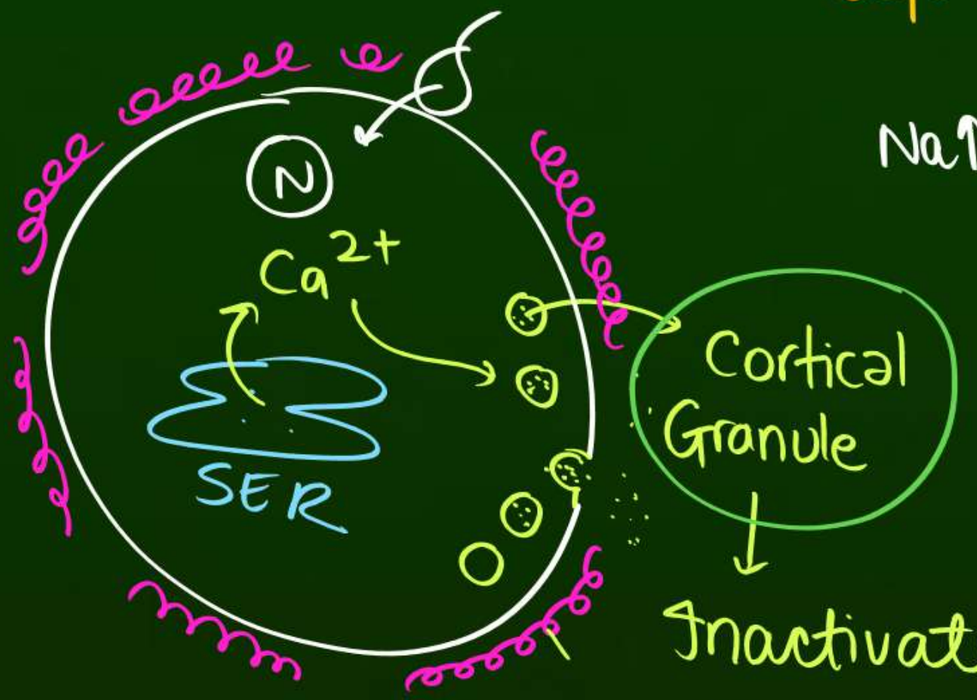
① → Fast block ~ 10 Sec

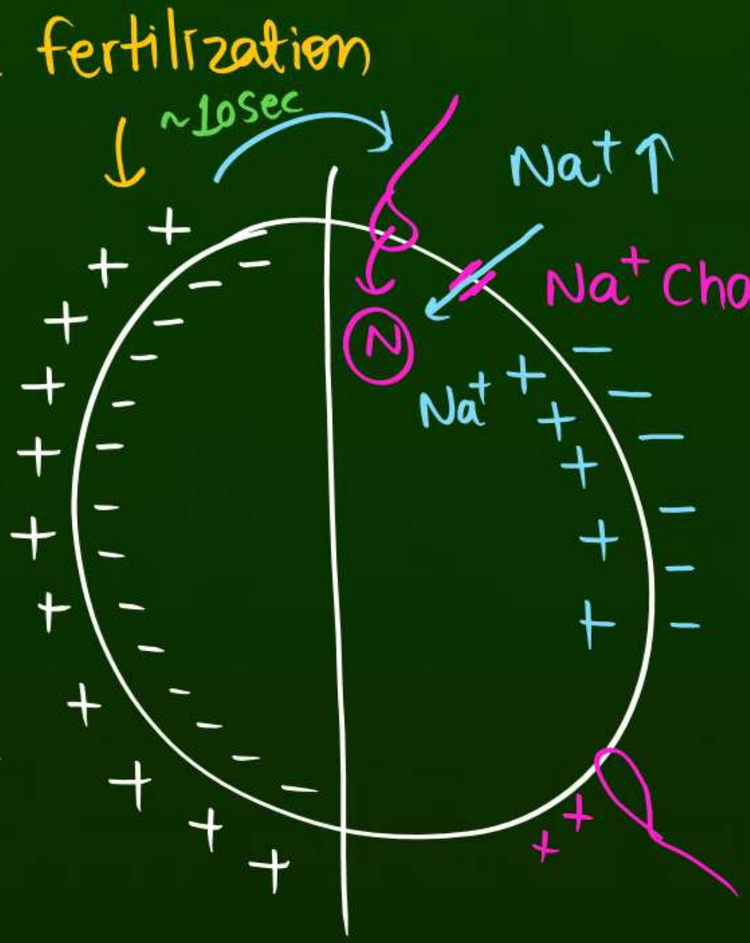
② → Slow block ~ 60 Sec

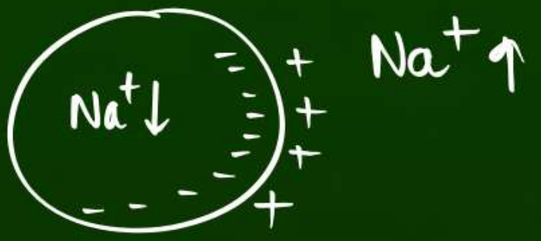
Cortical granule Reaction

Slow block : 60 Sec

fast block

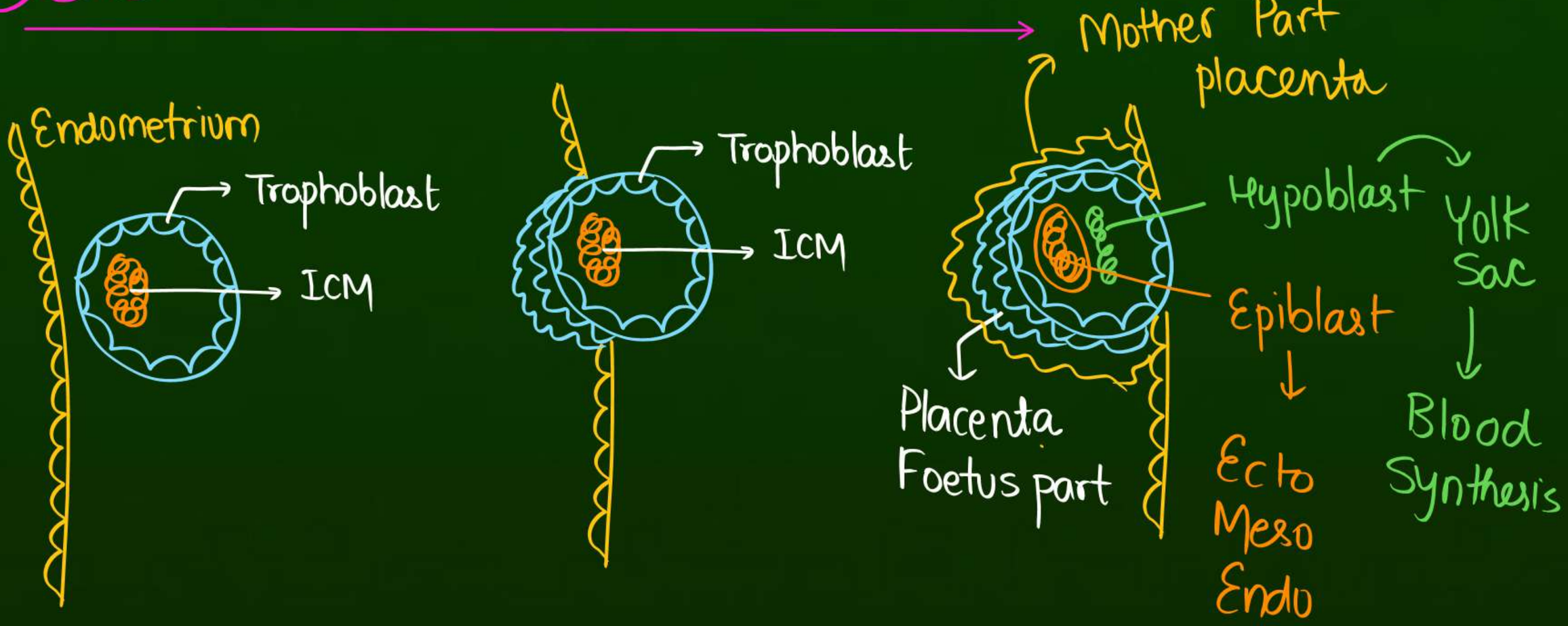






Implantation

"PREGNENCY"



Aminocentiosis

Genetic disorder ← Study → (~1L) * Amniotic fluid

baby ← * Amniotic mem

* Trophoblast → attach to Endometrium → Placenta ↑

* ICM : Inner Cell mass → Epiblast → Aminotic Ectoderm

Epiblast → Hypoblast → Yolk sac

Embryonic Epiblast → Ecto, Meso, Endo







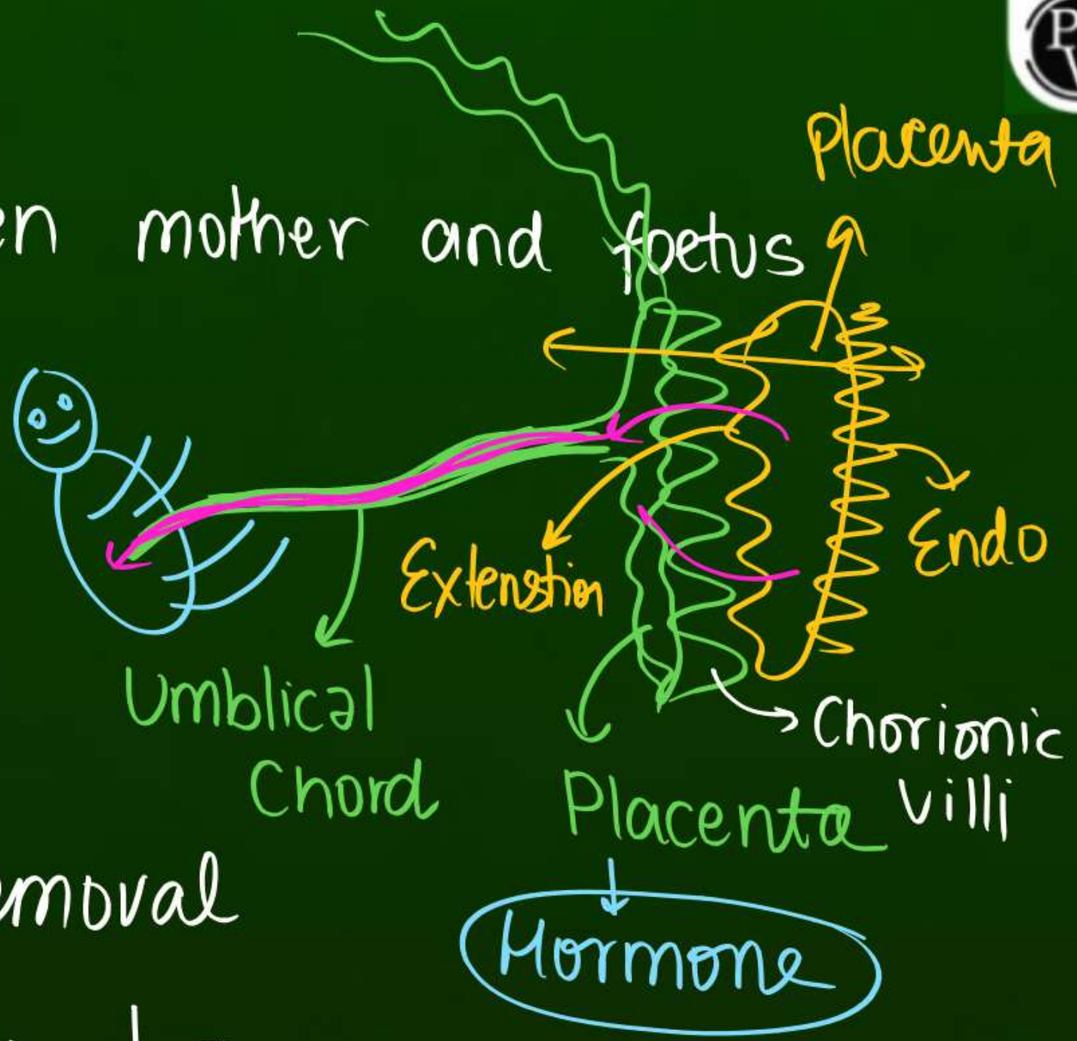
Embronic Epiblast

Aminotic Ectodurm

After attachment of trophoblast, Uterine cell rapidly divide and cover complete blastocyst, and blastocyst Embede in Endometrium Called Implantation that lead to pregnency

Placenta Connection between mother and foetus

1. Diffusion of nutrients
2. Diffusion of O_2
3. Exchange of Gases
4. Nitrogenous waste removal
5. Passage of Antibodies | Immunoglobulin Ig

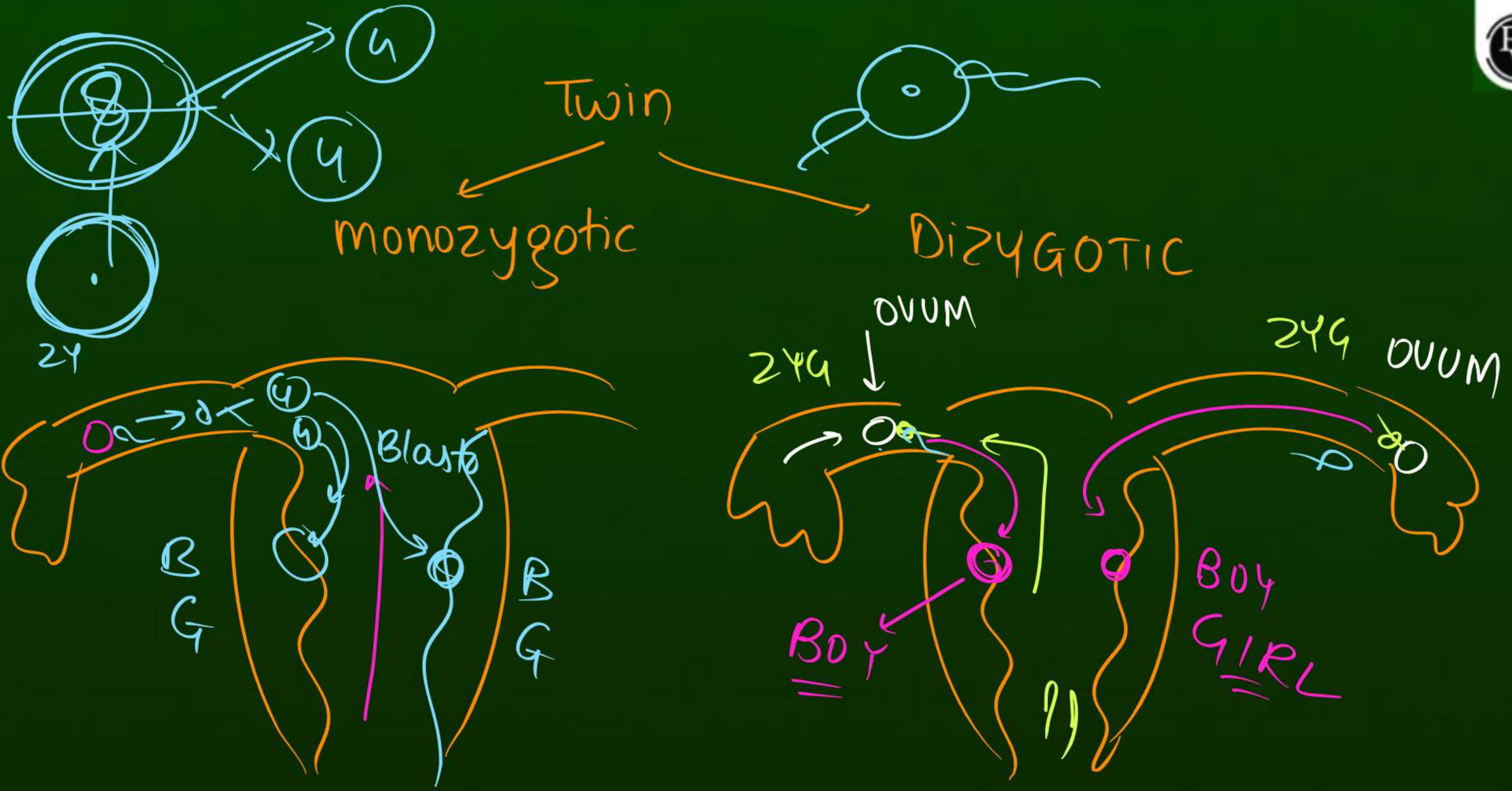


Exocrine → Enz

Placenta : Hormon | Endocrine Gland

1. hCG : act as LH during pregnancy
2. hPL : Human placental lactogen :- Milk Synthesis
3. Relexin :- Widen pubic Symphysis
4. Estrogen
5. Progestron

→ Maintain


Twin
monozygotic
DIZYGOTIC
ZYG
OVUM
ZYG
OVUM
Blast
B
G
B
G
BOY
BOY
GIRL

9 MONTH

Trimester	Month	Week	EVENT
1	1	4	→ Heart formation (दिल दे दिया है।)
	2	8	→ Foetus dev. limb + digit (")
	3	12	→ all mayor organ + External Genitalia
2	4	16	→ first movement of baby, hair on head
	5	20	
	6	24	→ body Cover by fine hair, Eyelid Separate + Eyelesh
3	7	28	→ fully foetus develop.
	8	32	
	9	36	

THANK YOU